

Let us consider the following examples

Example- 1: *Apis indica*, *Apis dorsata*, *Apis mellifera* and *Apis florea*

In the above example '*indica*', '*dorsata*', '*mellifera*' and '*florea*' are different species belonging to the same genus called *Apis*.

Example- 2: *Pheretima posthuma*, *Periplaneta americana* and *Panthera leo*. In the example 2 the words '*posthuma*', '*americana*' and '*leo*' are names of different species belonging to different genera.

*Sometimes closely related species of a genus can interbreed, but they generally give rise to sterile offspring. A cross between a female donkey and a male horse gives rise to the sterile offspring called '*Hinny*' (a sterile hybrid).*

Dobzhansky introduced the concept of '**Mendelian Population**' while defining a species. A Mendelian population is a group of sexually reproducing individuals within which mating takes place. They share a 'common gene pool'. The members of a species show **assortative** (*preferential*) mating. Populations of a species inhabiting different geographical areas are in a continuous process of adapting to the conditions of their surrounding environments. This leads to the evolution of new species, in course of time. Thus, species is **dynamic**.

Find out the scientific names of at least five animals around your home.

1.9 KINGDOM: ANIMALIA

Animalia includes eukaryotic, multicellular heterotrophs (generally obtain nourishment by ingestion i.e, holozoic method and digesting the ingested food). They have specific body plans and do not possess cell walls and photosynthetic pigments. Tissue formation is common except in the sponges. Higher forms show elaborate sensory and neuromotor mechanisms. Organ and organ-systems are developed in most of the groups. Nutrition is **holozoic** and a few are parasitic. Reserve food is mostly in the form of **glycogen**. These are mostly motile and move in search of food (except the sedentary forms). Muscle cells, sensory cells and nerve cells are present. Reproduction is mostly by sexual method; some reproduce by asexual



methods. Animalia includes simple sedentary organisms such as sponges to the highly evolved vertebrates, the mammals.

Classification of Kingdom Animalia

Kingdom Animalia is divided into two subkingdoms based on the organization of cells into tissues or higher levels of organisation.

- i) **Parazoa:** These are multicellular animals without the formation of well defined tissues. The only phylum included in this group is Porifera (e.g. *Sponges*).
- ii) **Eumetazoa:** These are multicellular animals with well defined tissues and higher levels of organization such as organs and organ- systems in the body. This group includes two taxonomic levels called Grades.

Grade:1. Radiata or Diploblastica

These are the first true metazoans or eumetazoans with radial symmetry and diploblastic body (having two primary germinal layers, the ectoderm and endoderm) Phyla : Cnidaria (e.g. *Hydra*, *Aurelia*, etc.) and Ctenophora (comb jellies) are included in this Grade.

Grade: 2. Bilateria or Triploblastica

This Grade includes eumetazoans with bilateral symmetry and triploblastic body (having three primary germinal layers, the ectoderm, mesoderm and endoderm). This includes two taxonomic levels called **Divisions**.

Division: I: Protostomia (*Proto* : first ; *Stomium* : mouth)

The eumetazoans in which blastopore develops into mouth are referred to as the protostomians . In the protostomes cleavage pattern is **spiral and determinate**. This Division includes three Sub-Divisions.

Sub-division: i) Acoelomata: (*a*:without; *coelom*: bodycavity; *ata*:having)

These organisms are the first triploblastic eumetazoans with **solid body plan** (without a body cavity/space between the body wall and visceral organs). The perivisceral space is filled with a tissue called parenchyma/ mesenchyme (derived from the germinal layer called mesoderm) e.g. Phylum Platyhelminthes (flat worms).

Sub-division: ii) Pseudocoelomata (*pseudo*:false; *coelom*:bodycavity; *ata*:having):

Animals of this group possess a body cavity between the body wall and the alimentary canal (perivisceral space derived from the embryonic blastocoel).

However it is not a 'true coelom/ secondary body cavity' as it is not lined by mesodermal epithelial layers (parietal and visceral peritoneal layers). Pseudocoel is a remnant of the embryonic blastocoel/ primary body cavity e.g., the phylum Nematoda. (Note: Truecoelom (eucoelom) is called secondary body cavity).

Sub-division: iii) Schizocoelomata: (*schizo*:splitting; *coelom*:body cavity; *ata*:having) They have a ‘true coelom’, which is a ‘schizocoel’. It is formed by **splitting** of the mesoderm into outer somatic and inner splanchnic layers e.g. the phyla Annelida, Arthropoda and Mollusca. (Note: In the vertebrates coelom is formed by the splitting of the mesoderm (called secondarily schizocoelic)).

Division: II) Deuterostomia (*deuteron*: secondary; *stomium*: mouth)

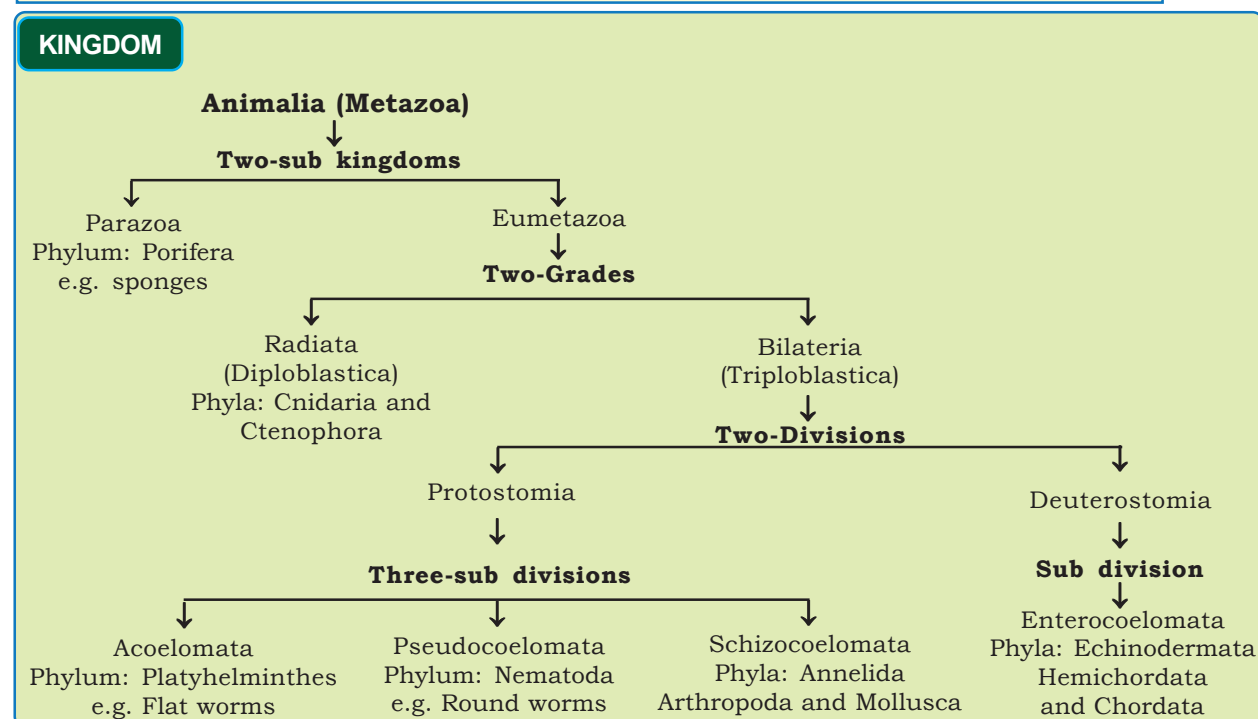
These are eucoelomates in which anus is formed from or near the blastopore. Mouth is formed later (secondarily), away from the blastopore at the opposite end. In deuterostomes cleavage pattern is ‘**radial**’ and ‘**indeterminate**’. It includes only one subdivision, **Enterocoelomata**.

Sub-division: Enterocoelomata

They have a true coelom, which is an ‘enterocoel’. It is formed by the out pouching of the archenteron. The phyla Echinodermata, Hemichordata and Chordata are included in this subdivision.

Schizocoelomates and enterocoelomates are together called ‘eucoelomates’ as they possess a ‘true coelom’.

Based on the knowledge gained, can you write the taxonomic position of human being (taxonomic hierarchy) starting from the Kingdom to the species/subspecies? (Take the help of your teacher, if you fail to do).



Note : Embryos of protostomes are called ‘**mosaic embryos**’ and those of deuterostomes are called ‘**regulative embryos**’